Propensity Scores

Lucy D'Agostino McGowan

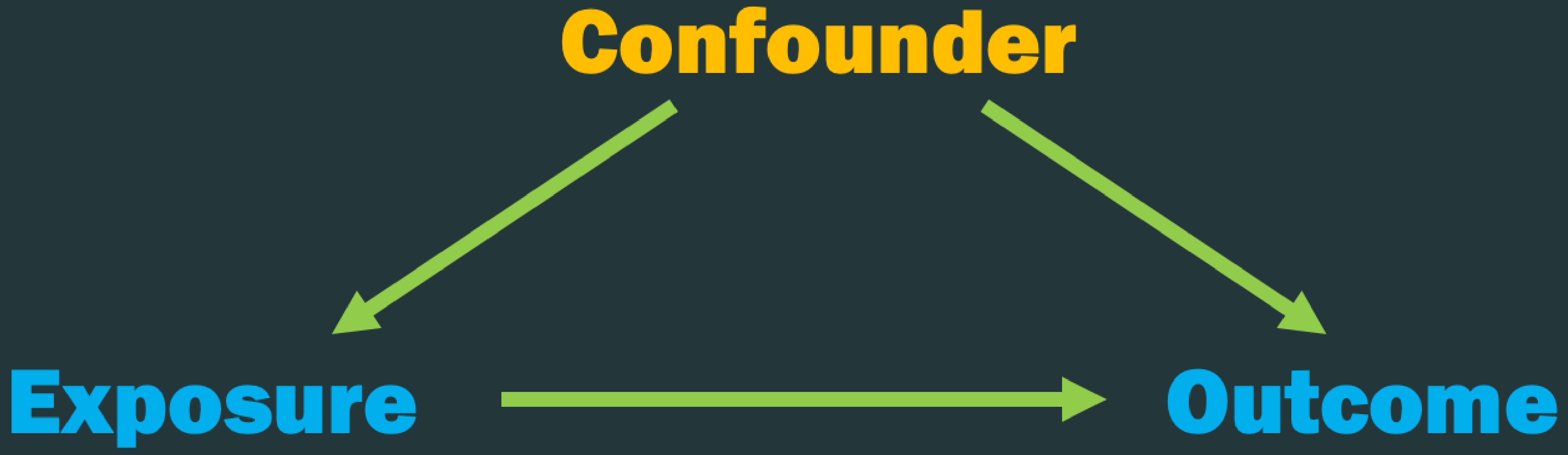
Wake Forest University

Observational Studies

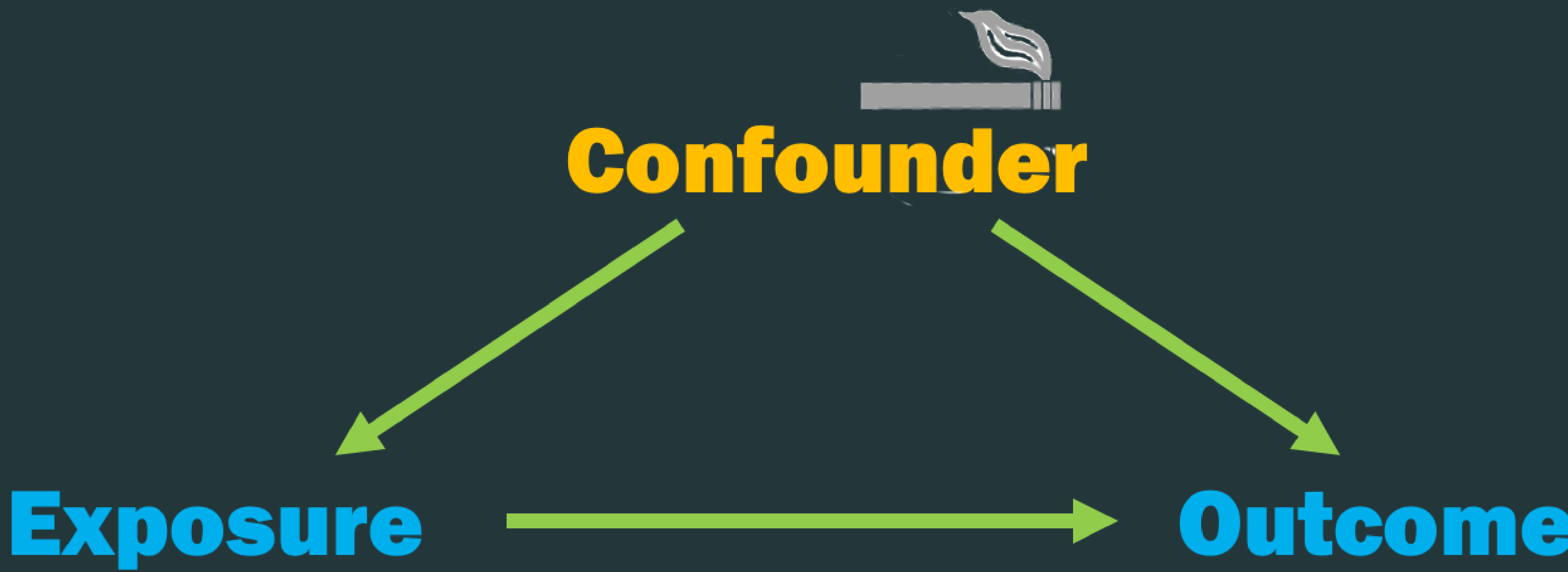
Treatment **P=0.9**

Control **P=0.1**

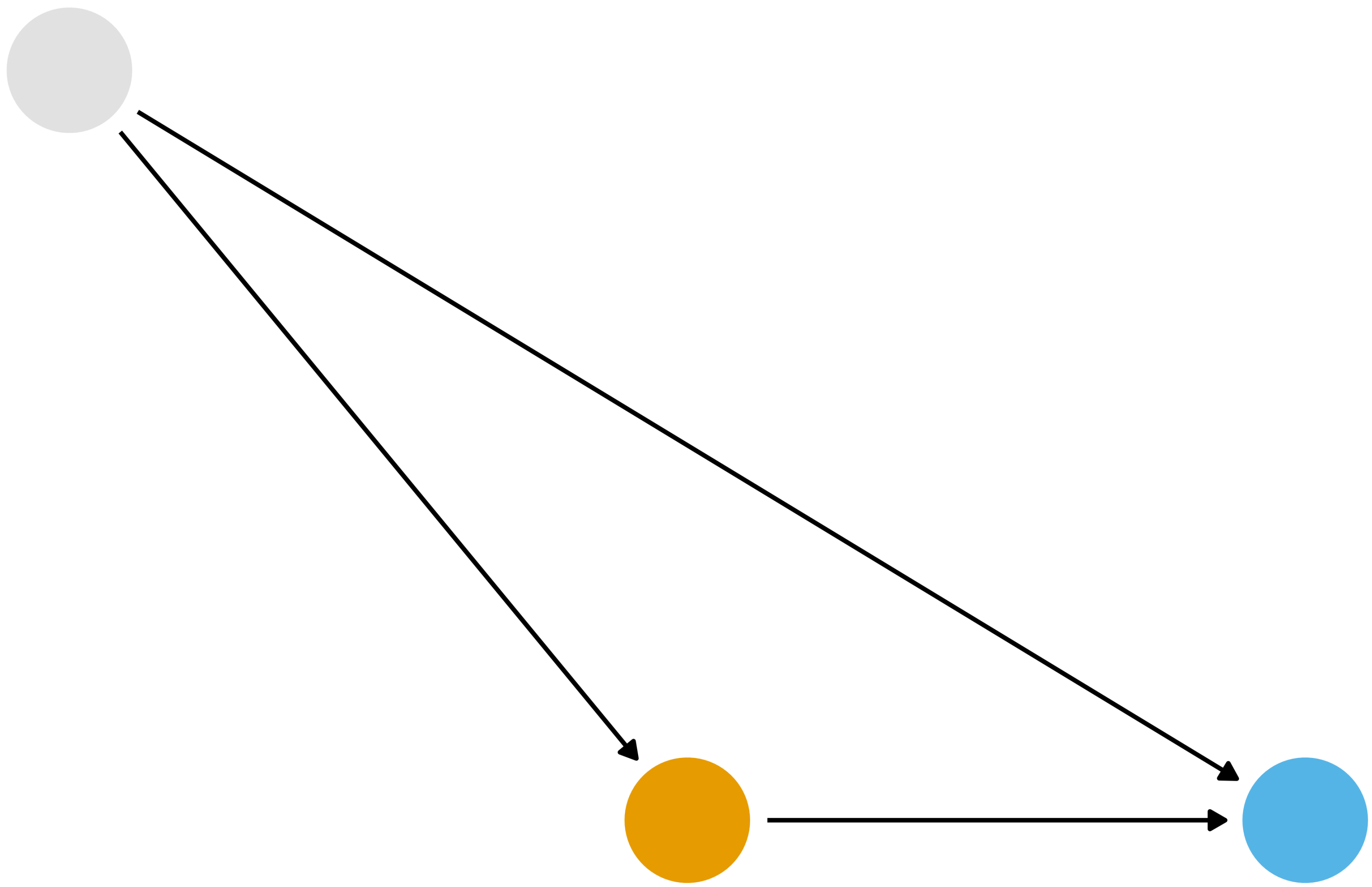
Confounding



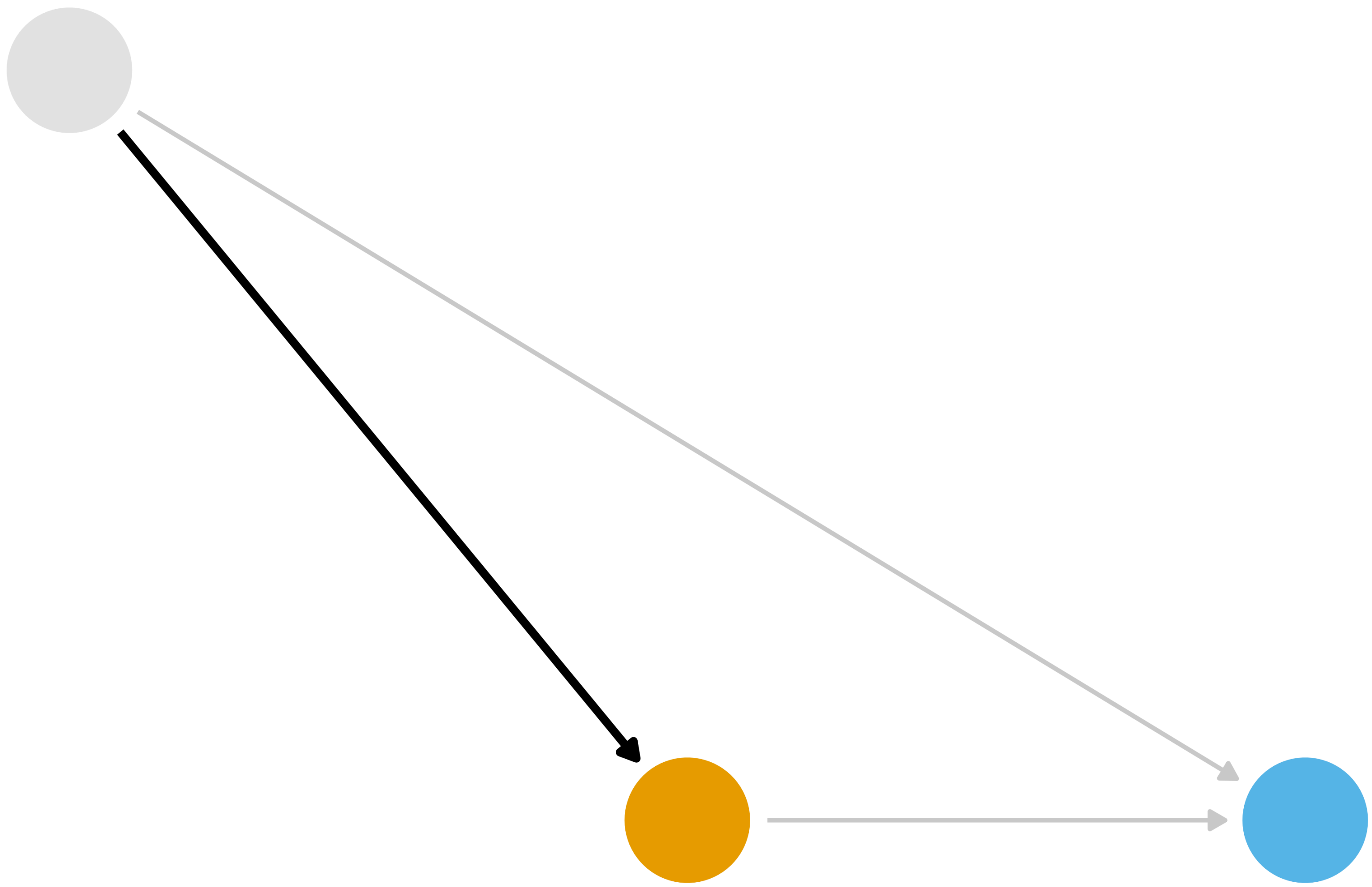
Confounding



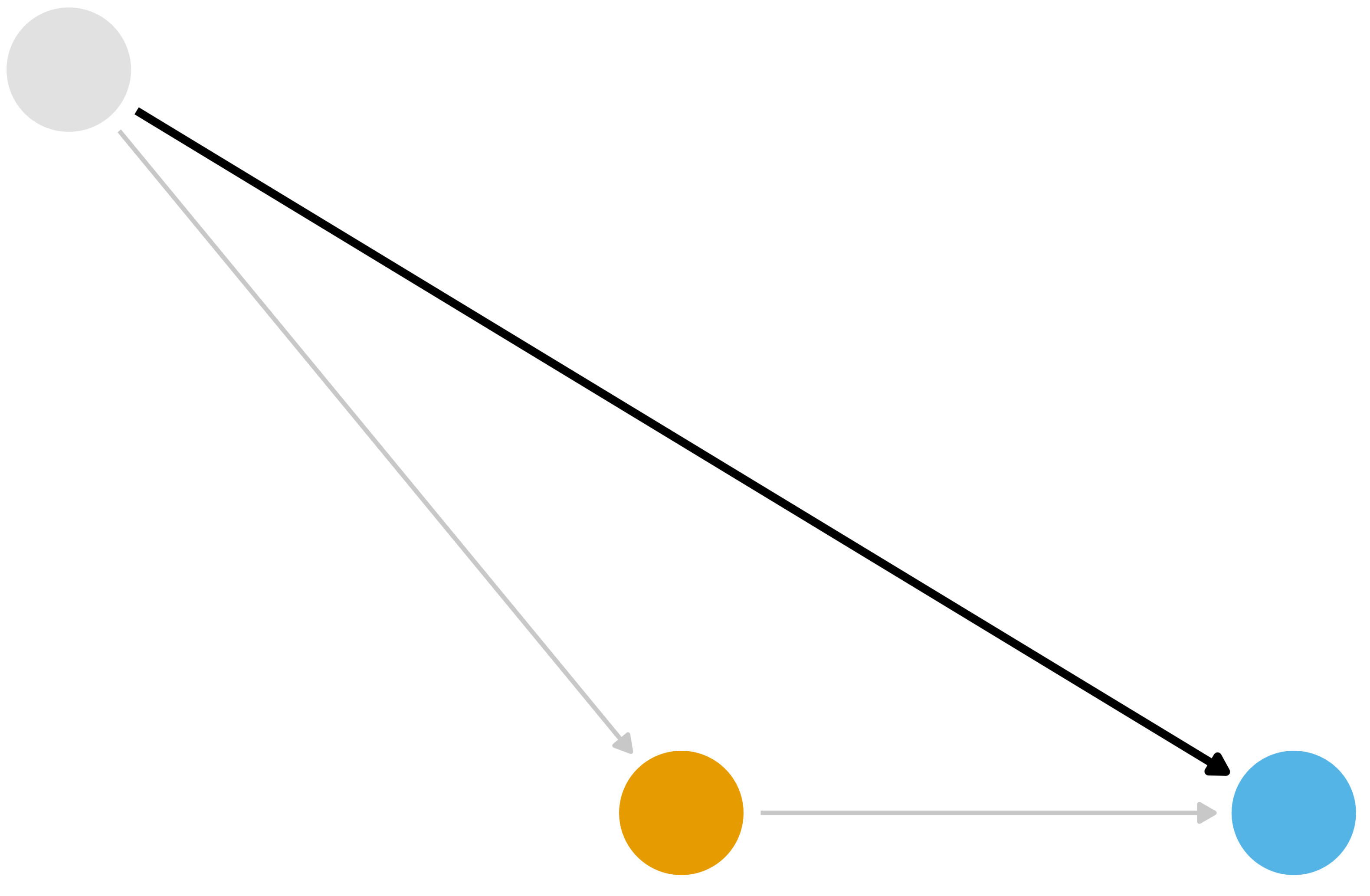
What part of the DAG do we want to try to deal with?



What part of the DAG do we want to try to deal with?



What part of the DAG do we want to try to deal with?



Which relationship should we model?

- Paths into the exposure: **propensity scores**
- Paths into the outcome: **outcome models**
- Both: **doubly robust** estimators

The hidden experiment

- Imagine the exposure had been randomized with a probability of 0.5

- Then the arrows into the exposure wouldn't exist

- In an experiment, the propensity score is *known*: `rbinom(1, 1, 0.5)`

The hidden experiment

- Now imagine randomizing the exposure *within levels of the confounders*
- Still a randomized experiment, but the probability of exposure depends on the covariates
- This gives us **conditional exchangeability**

The hidden experiment

- In observational data, these probabilities are not known
- But *something* determined who was exposed: what was that decision process?
- Domain knowledge and statistics can estimate it: the **propensity score**

Propensity scores

Rosenbaum and Rubin showed in observational studies, conditioning on **propensity scores** can lead to unbiased estimates of the exposure effect

1. There are no unmeasured confounders
2. Every subject has a nonzero probability of receiving either exposure

Propensity scores

- Fit a **logistic regression** predicting exposure using known covariates

- Each individuals' predicted values are the **propensity scores**

Propensity scores

```r
library(tidyverse)
library(broom)
library(causalworkshop)
library(halfmoon)
```

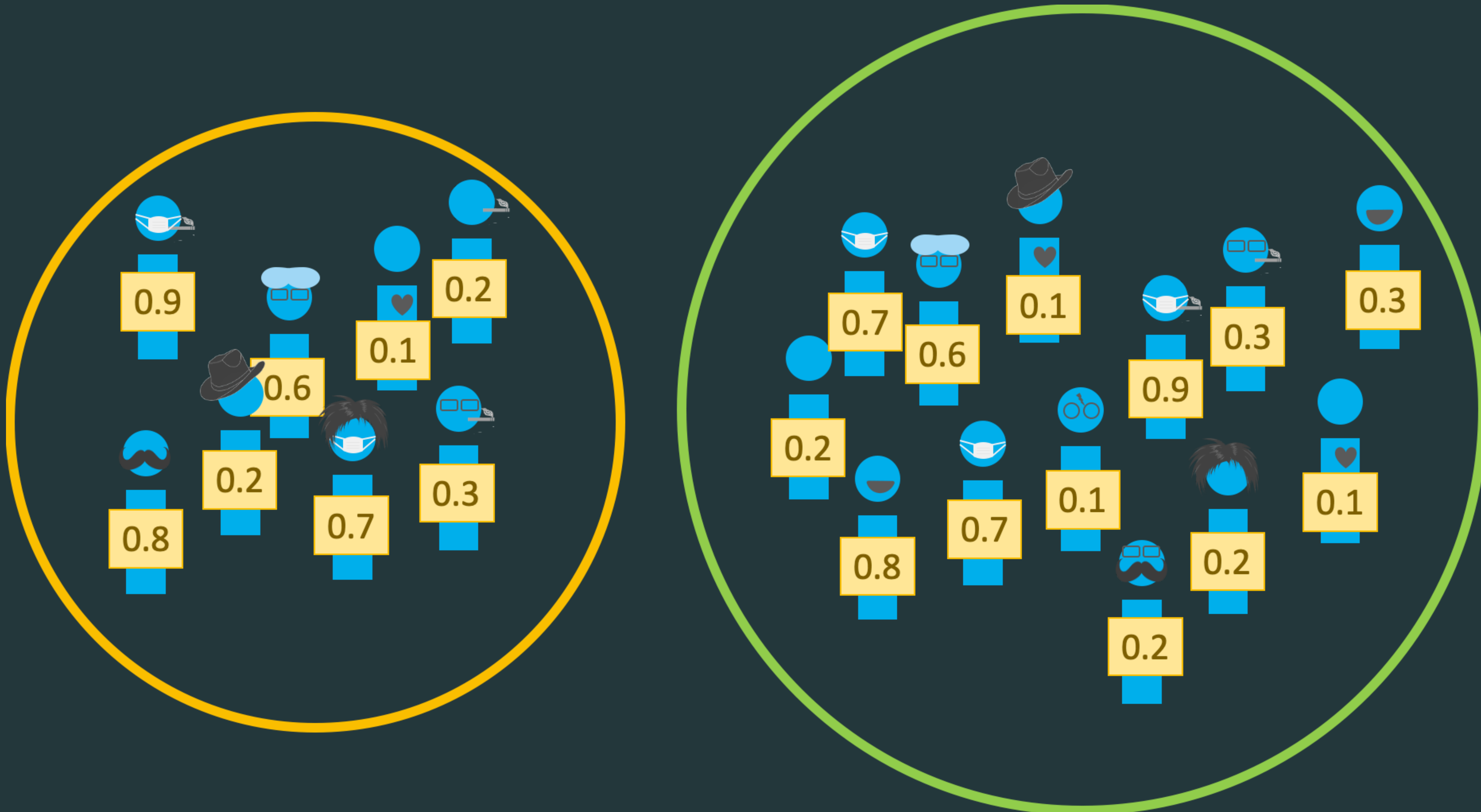
Propensity scores

```r
glm(
  exposure ~ confounder_1 + confounder_2 + confounder_3 + ...,
  data = df,
  family = binomial()
)
```

Propensity scores

```
glm(
  exposure ~ confounder_1 + confounder_2 + confounder_3 + ...,
  data = df,
  family = binomial()
) |>
  augment(type.predict = "response", data = df)
```

Propensity scores



Propensity scores

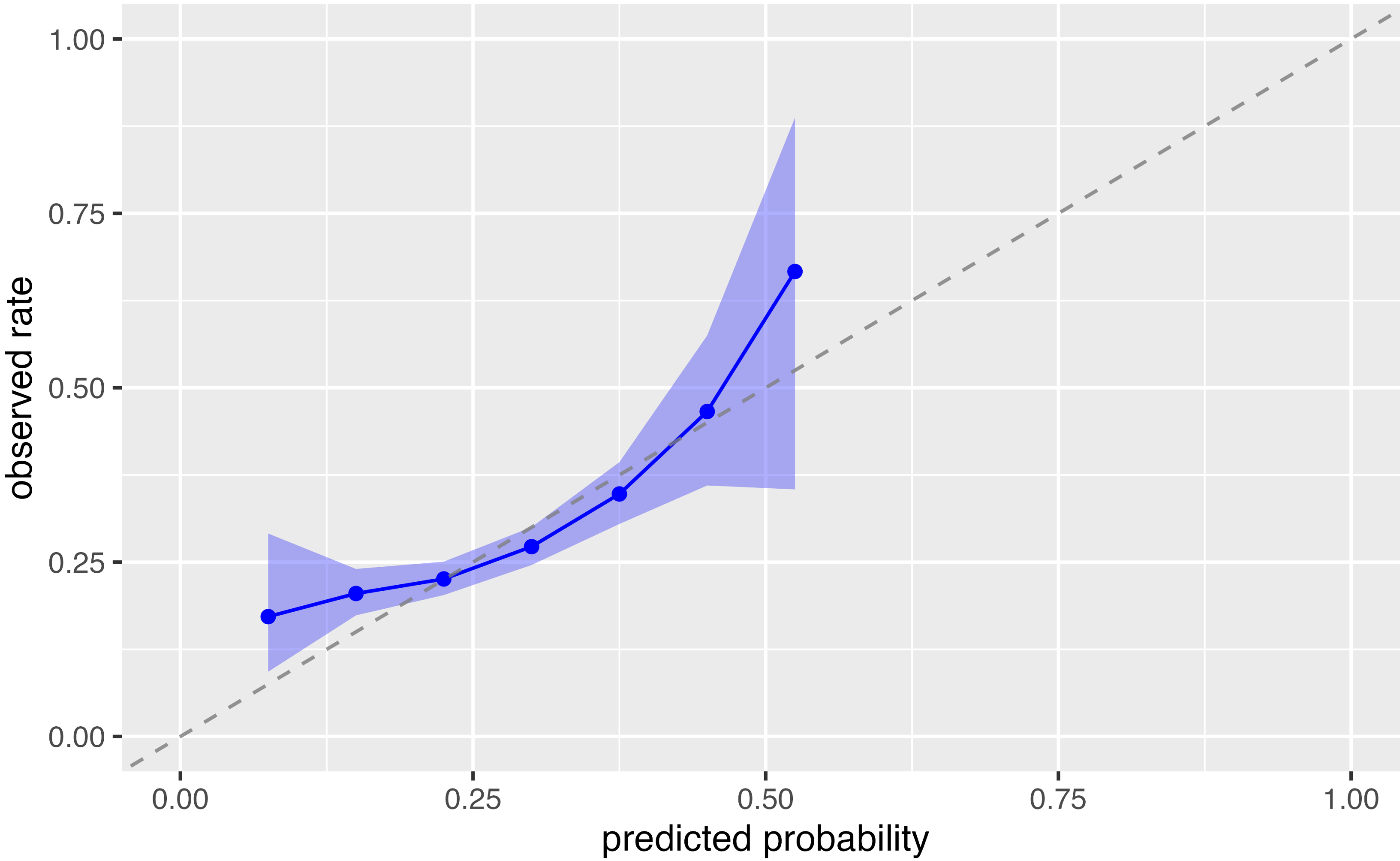
```r
propensity_model <- glm(
  net ~ income + health + temperature,
  data = net_data,
  family = binomial()
)

net_data_ps <- propensity_model |>
  augment(type.predict = "response", data = net_data)
```

Calibration

If we look at all the households with a predicted probability of 0.3, about 30% of them should use a net

```r
plot_model_calibration(
  propensity_model,
  method = "windowed",
  window_size = 0.15
)
```

Calibration

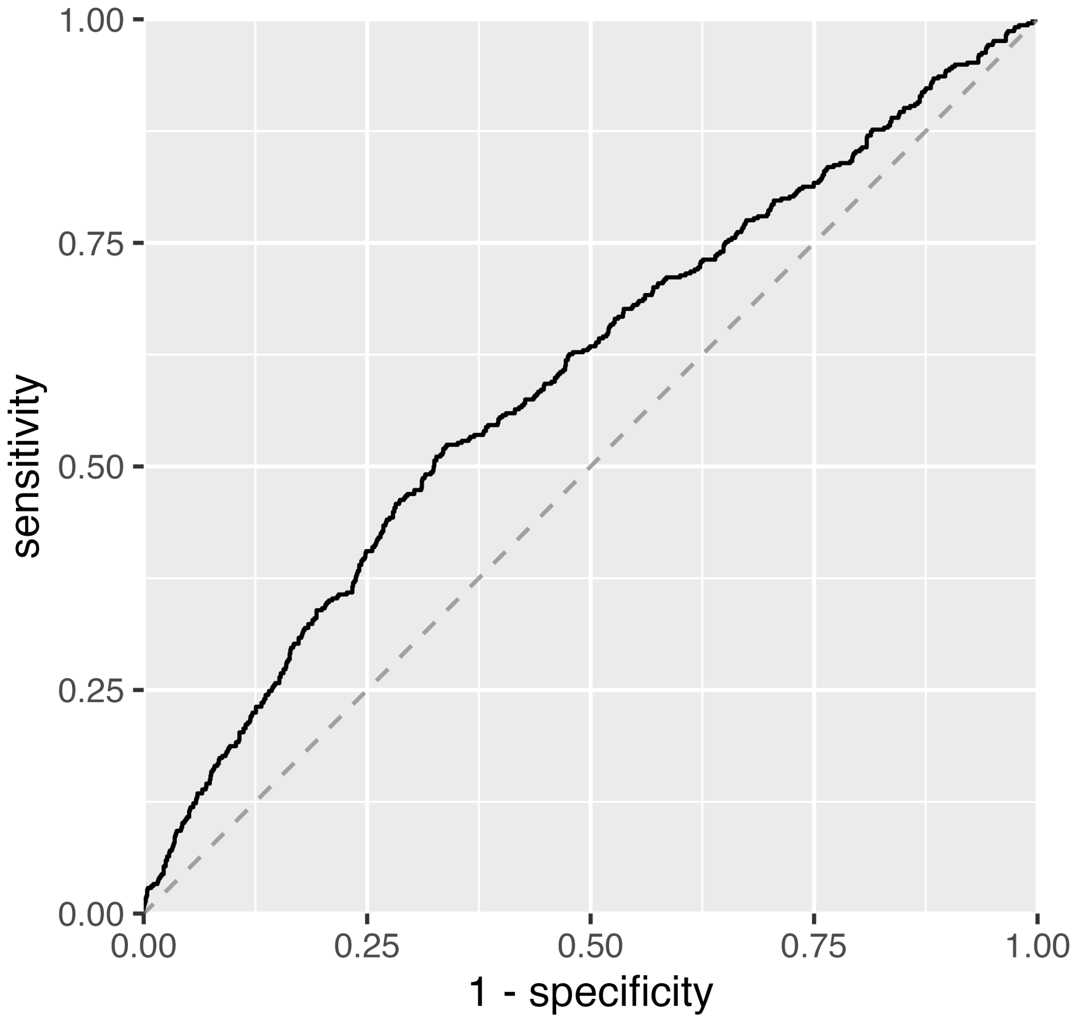


ROC curve

How well does the model discriminate between the exposed and the unexposed?

```
net_data_ps |>
  check_model_roc_curve(net, .fitted) |>
  plot_model_roc_curve()
```

ROC curve


1.00
0.75
0.50
0.25
0.00
sensitivity
0.00
0.25
0.50
0.75
1.00
1 - specificity

AUC

```r
check_model_auc(net_data_ps, .exposure = net, .fitted = .fitted)
```

```
# A tibble: 1 × 2
  method     auc
  <chr>    <dbl>
1 observed 0.600
```

- Perfect discrimination would be a violation of **positivity**

- Values between 0.6 and 0.8 usually mean we've picked up enough signal to adjust for

AUC

- There is no clear cutoff for a “good” AUC in causal inference
- A randomized trial has an expected AUC of 0.5
- We are not trying to optimize the AUC; we use it as a **diagnostic**

Example: The Seven Dwarfs Mine Train

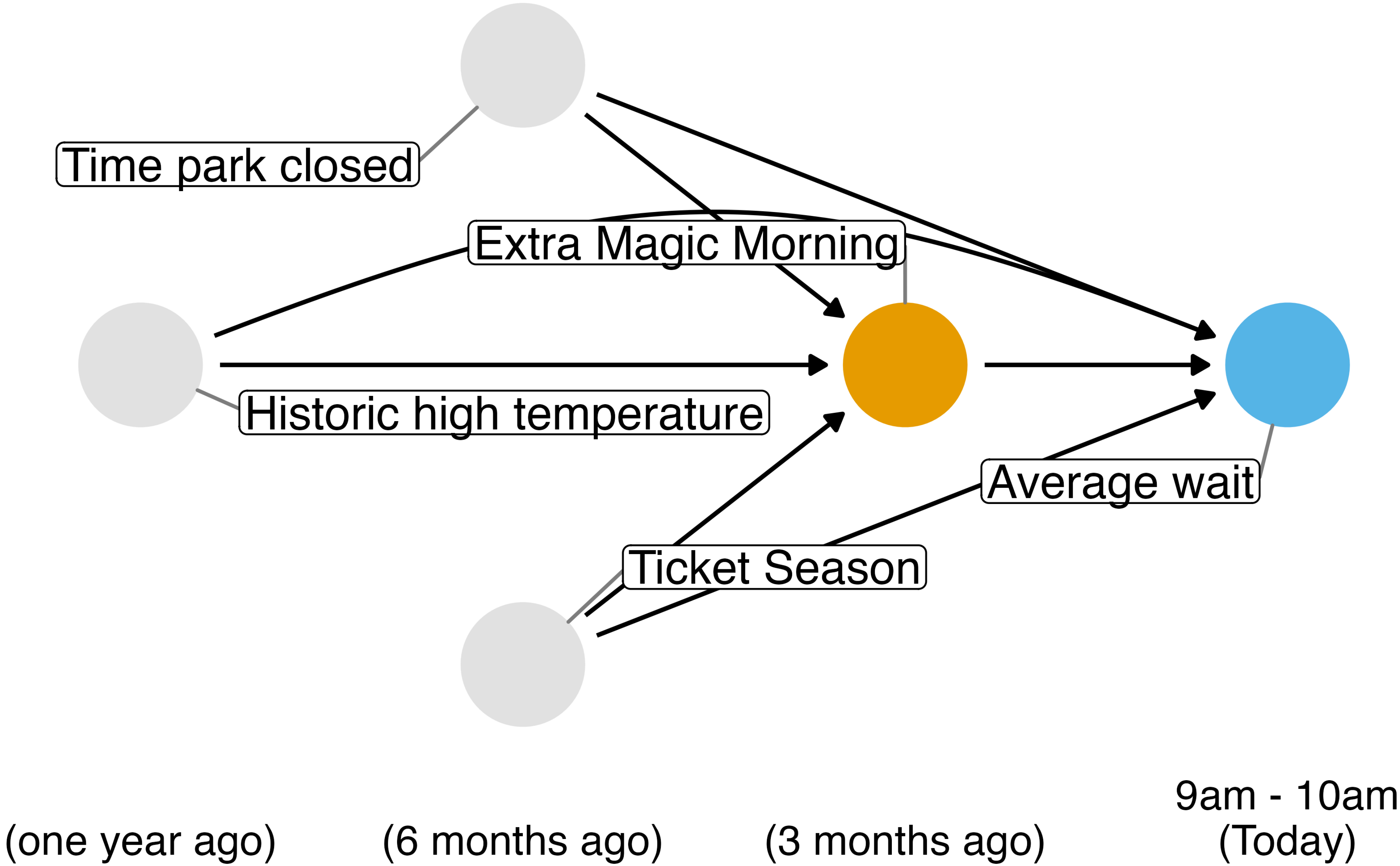
Photo by Anna CC-BY-SA-4.0

Historically, guests who stayed in a Walt Disney World resort hotel were able to access the park during “Extra Magic Hours” during which the park was closed to all other guests.

These extra hours could be in the morning or evening.

The Seven Dwarfs Mine Train is a ride at Walt Disney World’s Magic Kingdom. Typically, each day Magic Kingdom may or may not be selected to have these “Extra Magic Hours”.

We are interested in examining the relationship between whether there were **"Extra Magic Hours"** *in the morning and* **the average posted wait time** *for the Seven Dwarfs Mine Train the same day between 9 AM and 10 AM in 2018.*


Time park closed
Extra Magic Morning
Historic high temperature
Average wait
Ticket Season
9am - 10am
(one year ago)
(6 months ago)
(3 months ago)
(Today)

Your turn

Using the confounders identified in the previous DAG, fit a propensity score model for `park_extra_magic_morning`

Check the calibration, ROC curve, and AUC of your model

***Stretch*: Create two histograms, one of the propensity scores for days with extra morning magic hours and one for those without**

08:00